Use Case 9: Continuous Deployment

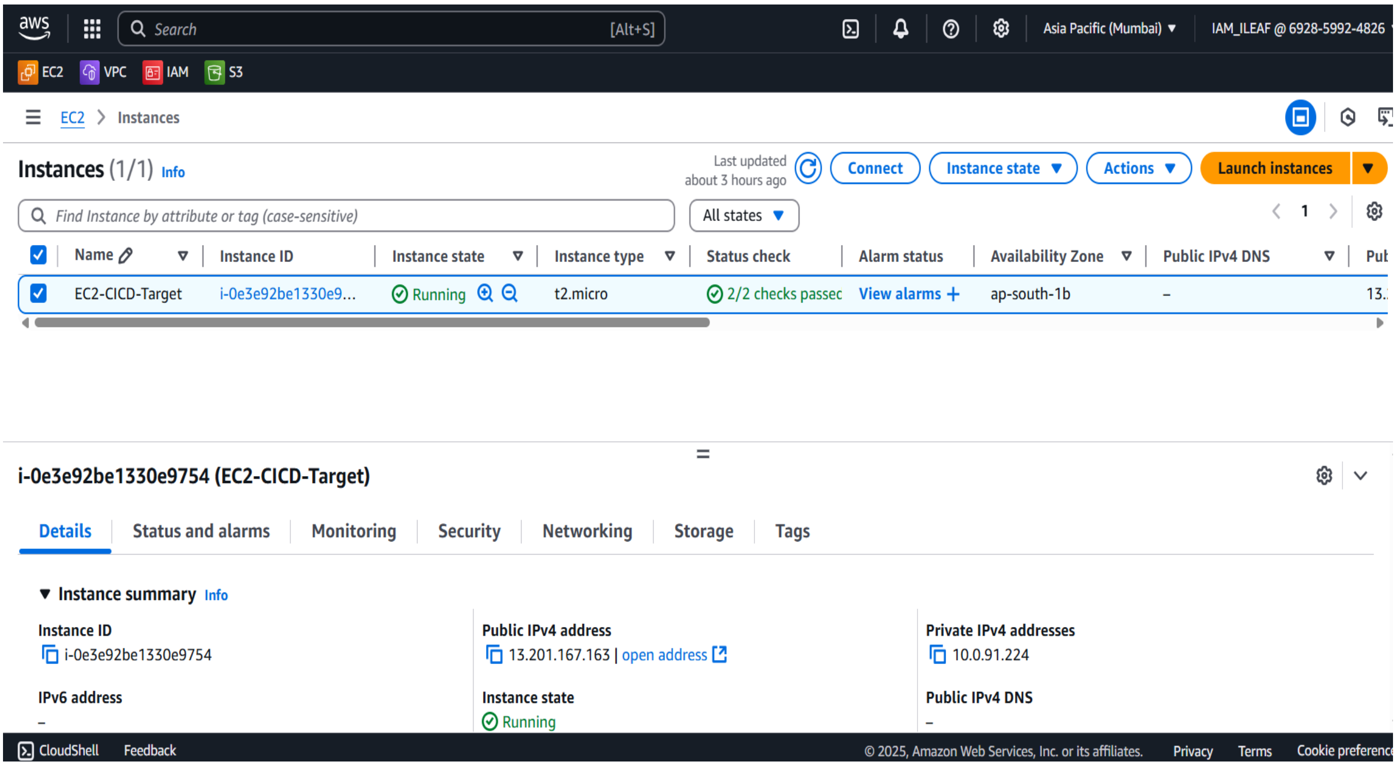
Scenario: Set up a CI/CD pipeline.

Tasks:

- Use CodePipeline, CodeBuild, and CodeDeploy.
- Deploy updates to an application hosted on EC2.
- Rollback automatically on deployment failures.

Step 1: Set Up EC2 Environment

1. Launch EC2 Instances:
    - Go to the AWS Management Console and launch one or more


aws
Search
[Alt+S]
Asia Pacific (Mumbai)
IAM_ILEAF @ 6928-5992-4826
EC2
VPC
IAM
S3
EC2 > Instances
Instances (1/1) Info
Last updated about 3 hours ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
1
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
EC2-CICD-Target
i-0e3e92be1330e9...
Running
t2.micro
2/2 checks passec
View alarms +
ap-south-1b
–
i-0e3e92be1330e9754 (EC2-CICD-Target)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0e3e92be1330e9754
Public IPv4 address
13.201.167.163 | open address
Private IPv4 addresses
10.0.91.224
IPv6 address
–
Instance state
Running
Public IPv4 DNS
–
CloudShell
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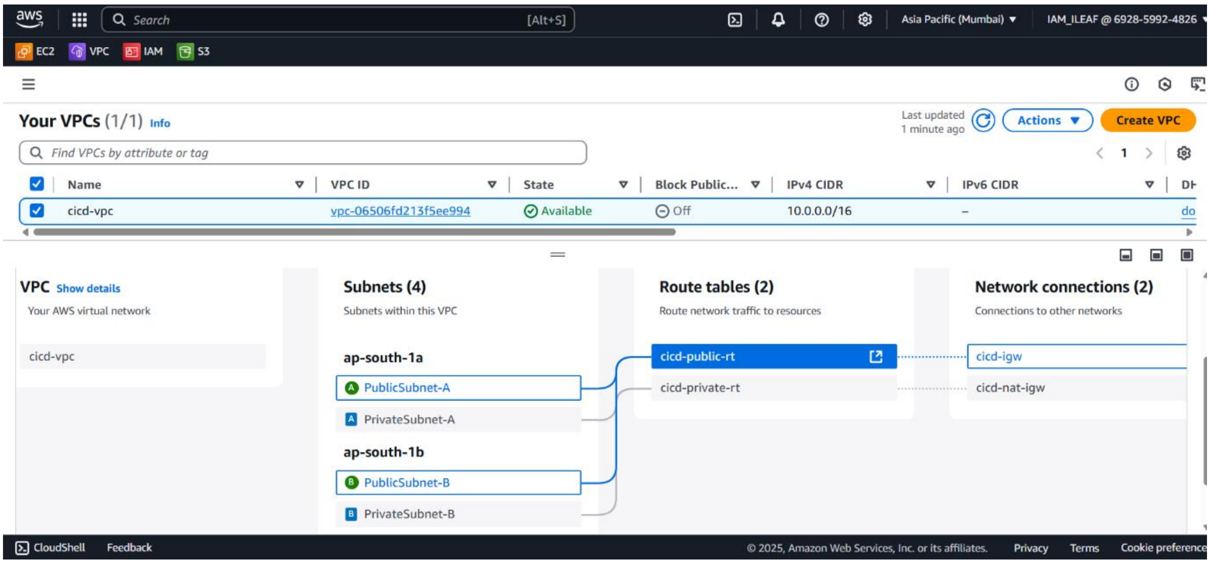
- EC2 instances.
- Choose an Amazon Machine Image (AMI) that suits your application, such as Amazon Linux 2.
- Ensure the EC2 instances are in the same VPC and region.

2. Install Required Software on EC2:
   - Install dependencies required for your application, such as Python, Node.js, or Java.
   - Install and configure the CodeDeploy agent:

bash

CopyEdit

```
sudo yum update -y
sudo yum install ruby -y
sudo yum ins
```


aws
Search
[Alt+S]
Asia Pacific (Mumbai)
IAM_ILEAF @ 6928-5992-4826
EC2
VPC
IAM
S3
Your VPCs (1/1) Info
Last updated 1 minute ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Block Public...
IPv4 CIDR
IPv6 CIDR
cicd-vpc
vpc-06506fd213f5ee994
Available
Off
10.0.0.0/16
–
VPC Show details
Your AWS virtual network
cicd-vpc
Subnets (4)
Subnets within this VPC
ap-south-1a
PublicSubnet-A
PrivateSubnet-A
ap-south-1b
PublicSubnet-B
PrivateSubnet-B
Route tables (2)
Route network traffic to resources
cicd-public-rt
cicd-private-rt
Network connections (2)
Connections to other networks
cicd-igw
cicd-nat-igw
CloudShell
Feedback
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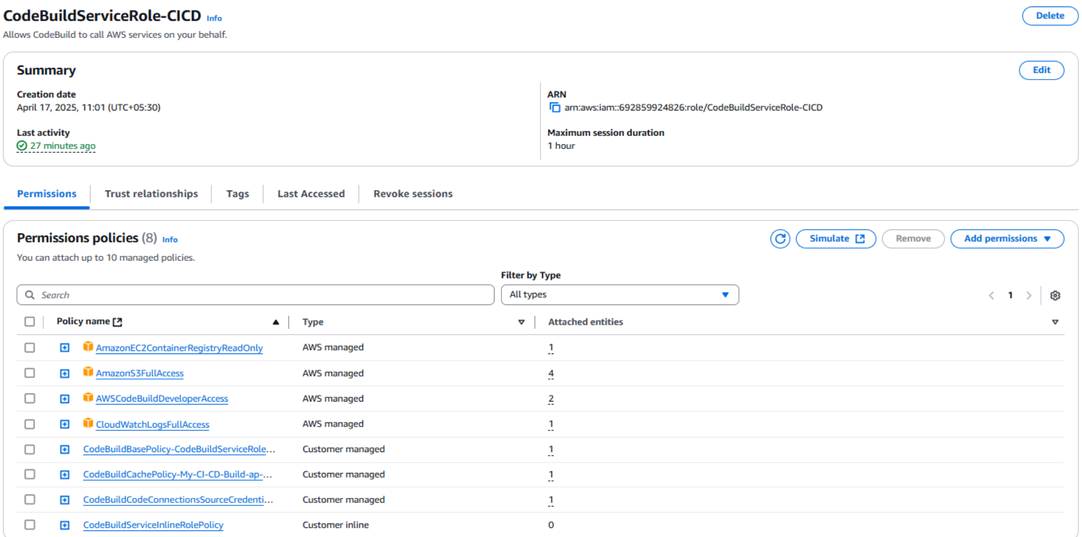
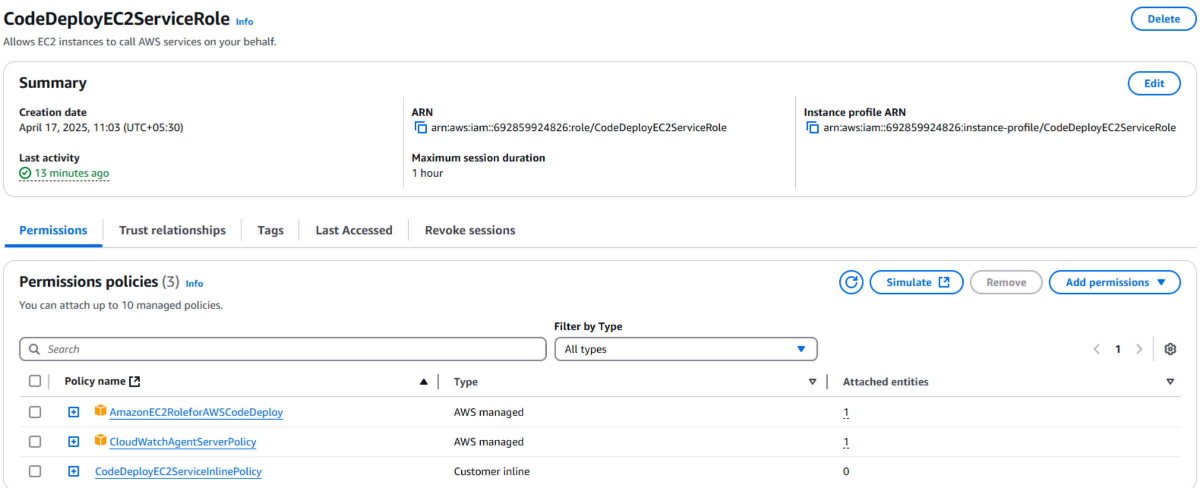

CodeBuildServiceRole-CICD Info
Delete
Allows CodeBuild to call AWS services on your behalf.
Summary
Edit
Creation date
April 17, 2025, 11:01 (UTC+05:30)
Last activity
27 minutes ago
ARN
arn:aws:iam::692859924826:role/CodeBuildServiceRole-CICD
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (8) Info
You can attach up to 10 managed policies.
Simulate
Remove
Add permissions
Filter by Type
Search
All types
Policy name
Type
Attached entities
AmazonEC2ContainerRegistryReadOnly AWS managed 1
AmazonS3FullAccess AWS managed 4
AWSCodeBuildDeveloperAccess AWS managed 2
CloudWatchLogsFullAccess AWS managed 1
CodeBuildBasePolicy-CodeBuildServiceRole... Customer managed 1
CodeBuildCachePolicy-My-CI-CD-Build-ap-... Customer managed 1
CodeBuildCodeConnectionsSourceCredenti... Customer managed 1
CodeBuildServiceInlineRolePolicy Customer inline 0


CodeDeployEC2ServiceRole Info
Delete
Allows EC2 instances to call AWS services on your behalf.
Summary
Edit
Creation date
April 17, 2025, 11:03 (UTC+05:30)
Last activity
13 minutes ago
ARN
arn:aws:iam::692859924826:role/CodeDeployEC2ServiceRole
Maximum session duration
1 hour
Instance profile ARN
arn:aws:iam::692859924826:instance-profile/CodeDeployEC2ServiceRole
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (3) Info
You can attach up to 10 managed policies.
Simulate
Remove
Add permissions
Filter by Type
Search
All types
Policy name
Type
Attached entities
AmazonEC2RoleforAWSCodeDeploy AWS managed 1
CloudWatchAgentServerPolicy AWS managed 1
CodeDeployEC2ServiceInlinePolicy Customer inline 0

tall wget -y

cd /home/ec2-user

wget https://bucket-name.s3.region.amazonaws.com/latest/install

chmod +x ./install

sudo ./install auto

sudo service codedeploy-agent start

3. Create an IAM Role for EC2:
    - Create an IAM Role with the AmazonEC2RoleforAWSCodeDeploy policy attached.
    - Attach this role to your EC2 instances.
4. Tag EC2 Instances:
    - Add a Tag to identify the EC2 instances to

CodeDeployServiceRole-CICD Info

Delete

Allows CodeDeploy to call AWS services such as Auto Scaling on your behalf.

Summary

Edit

Creation date
April 17, 2025, 11:04 (UTC+05:30)

Last activity
23 minutes ago

ARN
arn:aws:iam::692859924826:role/CodeDeployServiceRole-CICD

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (1) Info

Simulate | Remove | Add permissions

You can attach up to 10 managed policies.

Filter by Type

Search | All types

Policy name	Type	Attached entities
AWSCodeDeployRole	AWS managed	1

CodePipelineServiceRole Info

Delete

Allows EC2 instances to call AWS services on your behalf.

Summary

Edit

Creation date
April 17, 2025, 10:56 (UTC+05:30)

Last activity
24 minutes ago

ARN
arn:aws:iam::692859924826:role/CodePipelineServiceRole

Maximum session duration
1 hour

Instance profile ARN
arn:aws:iam::692859924826:instance-profile/CodePipelineServiceRole

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (5) Info

Simulate | Remove | Add permissions

You can attach up to 10 managed policies.

Filter by Type

Search | All types

Policy name	Type	Attached entities
AmazonS3FullAccess	AWS managed	4
AWSCodeBuildDeveloperAccess	AWS managed	2
AWSCodeDeployFullAccess	AWS managed	1
AWSCodePipeline_FullAccess	AWS managed	1
CodePipelineServiceInlinePolicy	Customer inline	0

deploy to, e.g.:

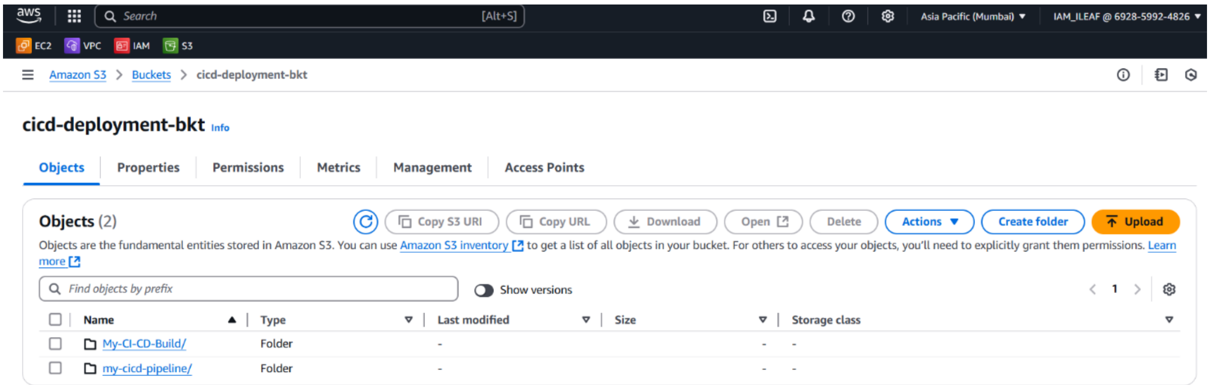
- Key: Name
- Value: CodeDeployTarget

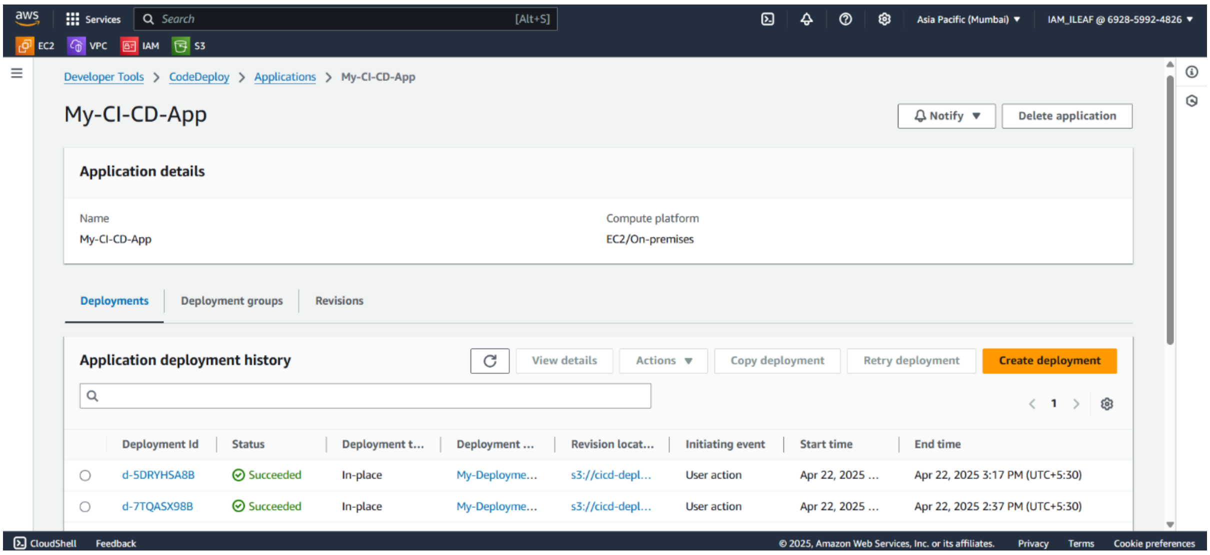
Step 2: Prepare Your Application

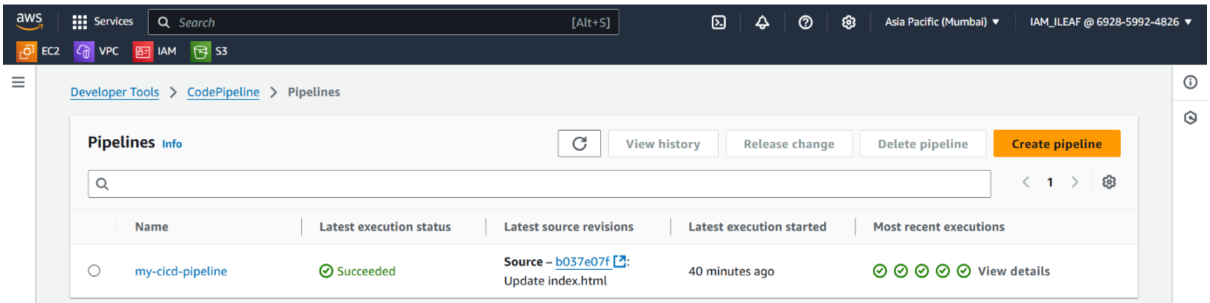
1. Package Your Application:
    - Organize the application files into a structure that includes an appspec.yml file.
1. Upload to S3 or CodeCommit:
    - Use an S3 bucket or AWS CodeCommit repository to store your application code.

Step 3: Set Up CodePipeline

1. Create a Pipeline:







Go to AWS CodePipeline and click Create Pipeline.

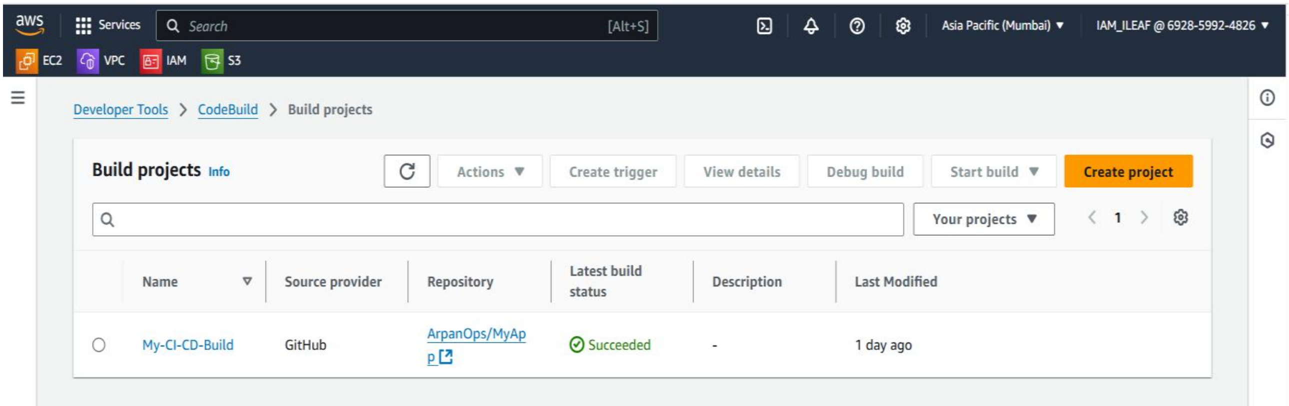
  - Enter a Pipeline Name (e.g., MyAppPipeline).
  - Choose a New Service Role to allow CodePipeline to access other AWS services.

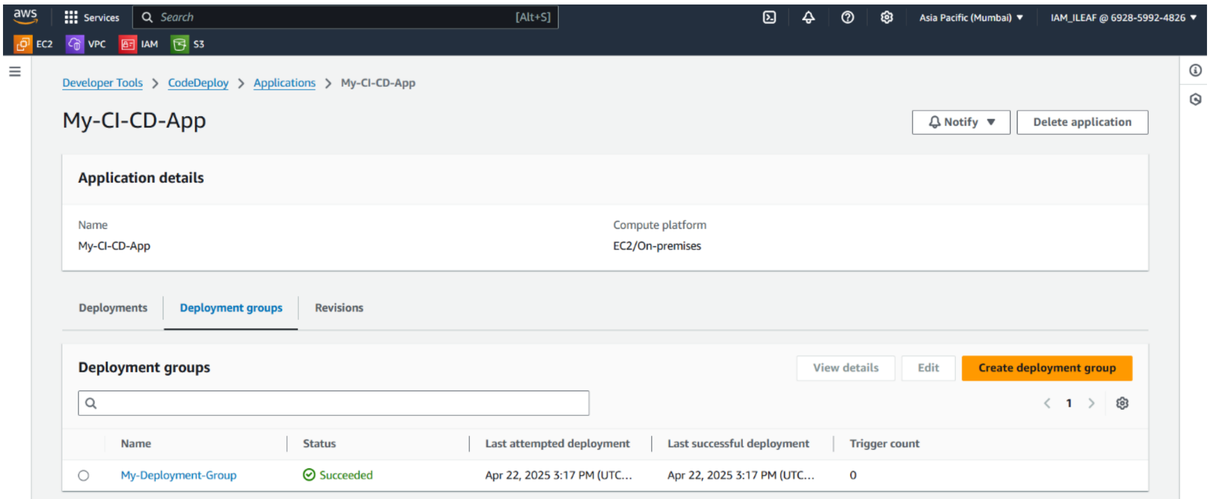
2. Add a Source Stage:
   - Choose the source for your code:
     - S3: If your application is stored in an S3 bucket, specify the bucket name and object key.
     - CodeCommit: If using CodeCommit, select the repository and branch.
3. Add a Build Stage:
   - Add AWS CodeBuild as the build provider.

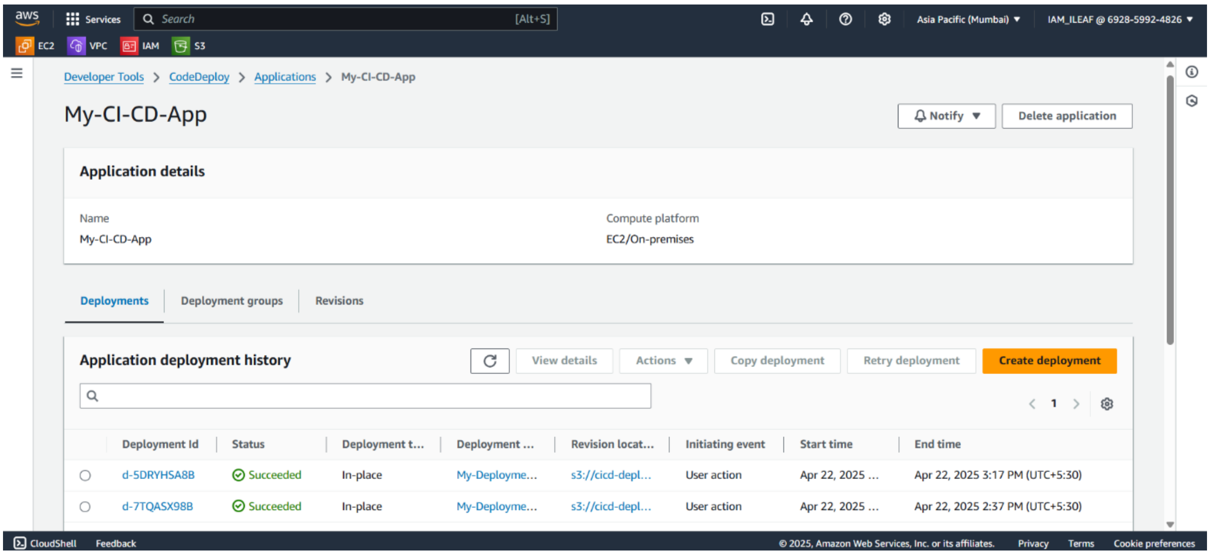
- Create a new CodeBuild project if not already set up.

Step 4: Configure CodeBuild

1. Create a CodeBuild Project:
    - In the CodeBuild Console, click Create Build Project.
    - Enter a name for the project (e.g., MyAppBuild).
    - Specify the source as CodePipeline.
    - Choose a Build Environment:
        - Operating System: Amazon Linux 2.
        - Runtime: Standard (choose the latest runtime version).
1. Test Your Build Project:


Services
Search [Alt+S]
Asia Pacific (Mumbai)
IAM_ILEAF @ 6928-5992-4826
EC2 VPC IAM S3
Developer Tools > CodeBuild > Build projects
Build projects Info
Actions
Create trigger
View details
Debug build
Start build
Create project
Your projects
1
Name
Source provider
Repository
Latest build status
Description
Last Modified
My-CI-CD-Build
GitHub
ArpanOps/MyApp
Succeeded
-
1 day ago


Services
Search [Alt+S]
Asia Pacific (Mumbai)
IAM_ILEAF @ 6928-5992-4826
EC2 VPC IAM S3
Developer Tools > CodeDeploy > Applications > My-CI-CD-App
My-CI-CD-App
Notify
Delete application
Application details
Name
My-CI-CD-App
Compute platform
EC2/On-premises
Deployments
Deployment groups
Revisions
Deployment groups
View details
Edit
Create deployment group
1
Name
Status
Last attempted deployment
Last successful deployment
Trigger count
My-Deployment-Group
Succeeded
Apr 22, 2025 3:17 PM (UTC...
Apr 22, 2025 3:17 PM (UTC...
0


Services
Search [Alt+S]
Asia Pacific (Mumbai)
IAM_ILEAF @ 6928-5992-4826
EC2 VPC IAM S3
Developer Tools > CodeDeploy > Applications > My-CI-CD-App
My-CI-CD-App
Notify
Delete application
Application details
Name
My-CI-CD-App
Compute platform
EC2/On-premises
Deployments
Deployment groups
Revisions
Application deployment history
View details
Actions
Copy deployment
Retry deployment
Create deployment
1
Deployment Id
Status
Deployment t...
Deployment ...
Revision locat...
Initiating event
Start time
End time
d-5DRYHSA8B
Succeeded
In-place
My-Deployme...
s3://cicd-depl...
User action
Apr 22, 2025 ...
Apr 22, 2025 3:17 PM (UTC+5:30)
d-7TQASX98B
Succeeded
In-place
My-Deployme...
s3://cicd-depl...
User action
Apr 22, 2025 ...
Apr 22, 2025 2:37 PM (UTC+5:30)
CloudShell
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2.

- Run a test build to ensure CodeBuild works correctly.

Step 5: Configure CodeDeploy

1. Create an Application in CodeDeploy:
   - Go to AWS CodeDeploy Console and create a new application.
   - Choose EC2/On-Premises as the compute platform.
   - Name your application (e.g., MyAppCodeDeploy).
2. Create a Deployment Group:
   - Specify a Deployment Group Name.
   - Select your application.

- In the Environment Configuration section, choose:
  - Amazon EC2 Instances.
  - Use the Tag Key-Value Pair to select your EC2 instances (e.g., Name=CodeDeployTarget).
- Enable Rollback on Deployment Failures.
- Specify a Service Role with the AWSCodeDeployRole policy.

3. Test CodeDeploy:
   - Perform a manual deployment to ensure that CodeDeploy can successfully deploy to your EC2 instances.

Step 6: Complete the Pipeline

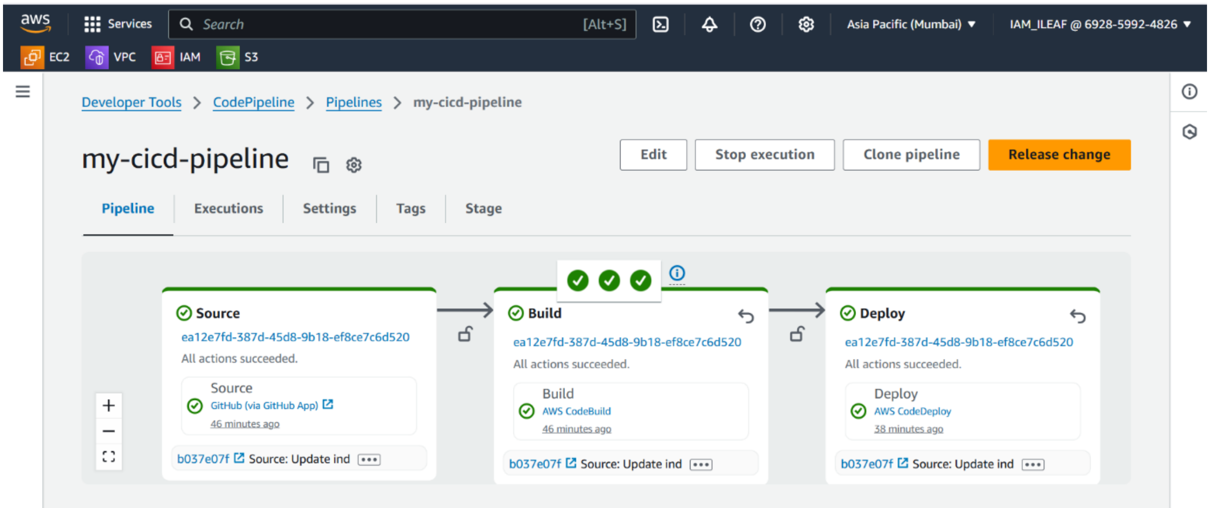
1. Add a Deploy Stage:

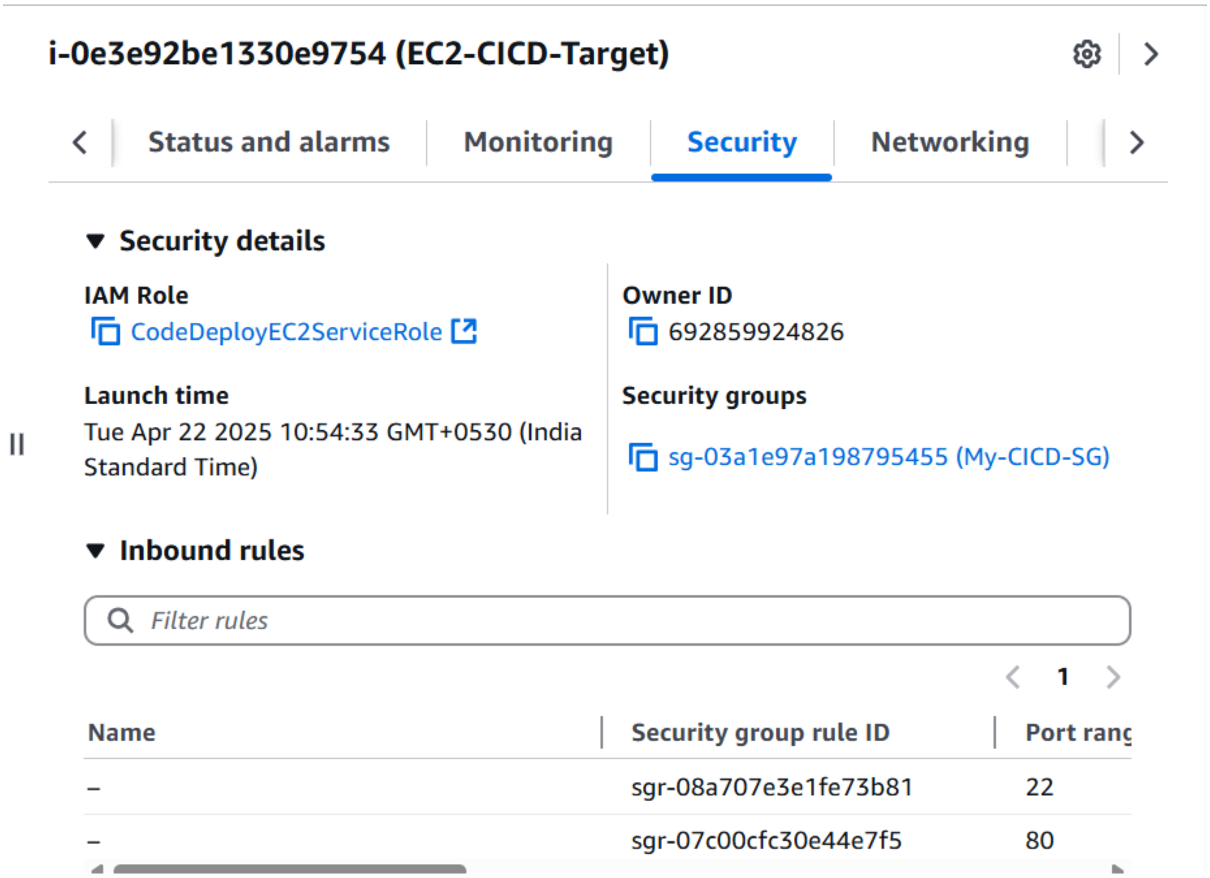
- Return to your CodePipeline configuration and add a Deploy Stage.
- Choose AWS CodeDeploy as the deploy provider.
- Select the application and deployment group created earlier.

2. Save and Start the Pipeline:
   - Save your pipeline and release a change to test the end-to-end process.

Step 7: Enable Automatic Rollbacks

1. Enable Rollbacks in CodeDeploy:
   - Go to your CodeDeploy Deployment Group.
   - Under Deployment Settings, enable Rollback on Deployment Failures.


Services
Search
[Alt+S]
Asia Pacific (Mumbai)
IAM_ILEAF @ 6928-5992-4826
EC2
VPC
IAM
S3
Developer Tools
CodePipeline
Pipelines
my-cicd-pipeline
my-cicd-pipeline
Edit
Stop execution
Clone pipeline
Release change
Pipeline
Executions
Settings
Tags
Stage
Source
ea12e7fd-387d-45d8-9b18-ef8ce7c6d520
All actions succeeded.
GitHub (via GitHub App)
46 minutes ago
b037e07f
Source: Update ind
Build
ea12e7fd-387d-45d8-9b18-ef8ce7c6d520
All actions succeeded.
AWS CodeBuild
46 minutes ago
b037e07f
Source: Update ind
Deploy
ea12e7fd-387d-45d8-9b18-ef8ce7c6d520
All actions succeeded.
AWS CodeDeploy
38 minutes ago
b037e07f
Source: Update ind


i-0e3e92be1330e9754 (EC2-CICD-Target)
Status and alarms
Monitoring
Security
Networking
Security details
IAM Role
CodeDeployEC2ServiceRole
Owner ID
692859924826
Launch time
Tue Apr 22 2025 10:54:33 GMT+0530 (India Standard Time)
Security groups
sg-03a1e97a198795455 (My-CICD-SG)
Inbound rules
Filter rules
1
Name
Security group rule ID
Port rang
–
sgr-08a707e3e1fe73b81
22
–
sgr-07c00cfc30e44e7f5
80

-
- CodeDeploy will monitor health checks and automatically roll back if an instance fails.

2. Configure Health Checks:
   - Define Health Check Scripts in your appspec.yml to verify that the application is running correctly.

Step 8: Test the Pipeline

1. Make a Code Change:
   - Push an update to your source repository or upload a new version to the S3 bucket.
2. Observe the Pipeline Execution:
   - Monitor the progress in the CodePipeline Console.

  - Check the Source, Build, and Deploy stages.
3. Simulate a Deployment Failure:
  - Introduce a failure (e.g., remove a required dependency) and observe the rollback process.

Step 9: Secure and Optimize

1. Secure IAM Roles:
  - Ensure that IAM roles have the principle of least privilege.
  - Limit access to only the resources needed for each stage of the pipeline.
2. Use AWS Secrets Manager:
  - Store sensitive information such as database credentials securely using Secrets Manager.

3. Cost Optimization:
    - Monitor costs using AWS Cost Explorer.
    - Use reserved or spot instances for EC2 where applicable.
    - Terminate unused resources.

Step 10: Monitor and Maintain

1. Set Up CloudWatch Alarms:
    - Monitor the health of your EC2 instances and pipeline stages.
    - Set up alarms for failed deployments or high latency.
2. Regularly Update Scripts and Configurations:
    - Update the appspec.yml and build scripts to adapt to changes in your application or environment.